VODAFONE | EFE | CCNA WORK PROJECT

TechLink Solutions

Multi-Branch Network Design

& Python Automation Document

Cairo ● Alexandria ● Giza | Static Routing + Python Automation

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1. Executive Summary

This document presents the full network design for **TechLink Solutions**, a company with three geographically distributed sites: **Head Office (HQ) in Cairo**, **Branch 1 in Alexandria**, and **Branch 2 in Giza**.

The design was implemented and tested in **Cisco Packet Tracer** and simulated via a **Python + JSON automation engine** that reads link status, detects failures, and automatically applies backup routing configurations without human intervention.

Feature	Implementation
Sites	Cairo HQ, Alexandria Branch 1, Giza Branch 2
Routing	Static routing with floating backup routes (AD=10)
Redundancy	Dual WAN links (Primary + Backup) per branch
Segmentation	VLAN 10 (Management) + VLAN 20 (Users) per site
Automation	Python script reading network_status.json
Simulation	Cisco Packet Tracer + Python JSON emulator

2. Network Topology Diagram

The topology uses a **Hub-and-Spoke** model. HQ Cairo is the central hub. Both Alexandria and Giza connect to HQ via a **primary Serial WAN link** and a **backup Serial WAN link**. Each site has a Cisco router (4331) connected to a Cisco Multilayer Switch (3650-24PS) which segments local traffic into two VLANs.

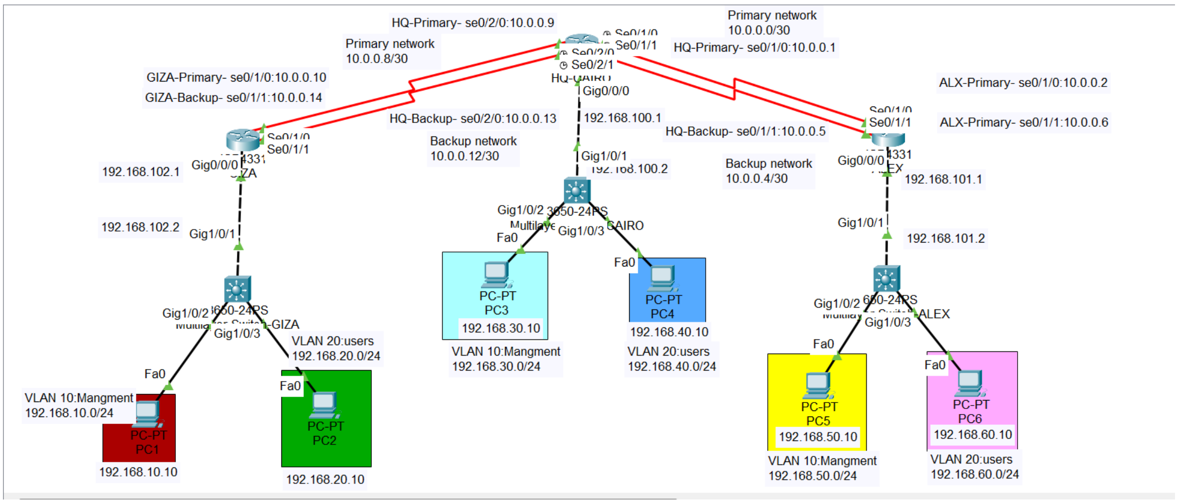

Figure 1 – TechLink Solutions Full Network Topology (Cisco Packet Tracer)

Red lines = WAN Serial links (Primary & Backup). Black dashed = Gigabit Ethernet (Router ↔ MLS). Solid black = Access

3. IP Addressing Plan

3.1 WAN Point-to-Point Links (/30)

Connection	Network	HQ IP	Branch IP	Purpose
HQ ↔ Alexandria Primary	10.0.0.0/30	10.0.0.1	10.0.0.2	Primary WAN
HQ ↔ Alexandria Backup	10.0.0.4/30	10.0.0.5	10.0.0.6	Backup WAN
HQ ↔ Giza Primary	10.0.0.8/30	10.0.0.9	10.0.0.10	Primary WAN
HQ ↔ Giza Backup	10.0.0.12/30	10.0.0.13	10.0.0.14	Backup WAN

3.2 Device Interface IP Table

Site	Device	Interface / VLAN	IP Address	Subnet Mask
Cairo HQ	HQ-CAIRO Router	`Gi0/0/0`	`192.168.100.1`	`255.255.255.252`
	MLS-CAIRO Switch	`Gi1/0/1`	`192.168.100.2`	`255.255.255.252`
	MLS-CAIRO VLAN 10 (Mgmt)	`SVI VLAN 10`	`192.168.30.1`	`255.255.255.0`
	PC3	`NIC`	`192.168.30.10`	`255.255.255.0`
	MLS-CAIRO VLAN 20 (Users)	`SVI VLAN 20`	`192.168.40.1`	`255.255.255.0`
	PC4	`NIC`	`192.168.40.10`	`255.255.255.0`
Alexandria	ALEX Router	`Gi0/0/0`	`192.168.101.1`	`255.255.255.252`
	MLS-ALEX Switch	`Gi1/0/1`	`192.168.101.2`	`255.255.255.252`
	MLS-ALEX VLAN 10 (Mgmt)	`SVI VLAN 10`	`192.168.50.1`	`255.255.255.0`
	PC5	`NIC`	`192.168.50.10`	`255.255.255.0`
	MLS-ALEX VLAN 20 (Users)	`SVI VLAN 20`	`192.168.60.1`	`255.255.255.0`
	PC6	`NIC`	`192.168.60.10`	`255.255.255.0`
Giza	GIZA Router	`Gi0/0/0`	`192.168.102.1`	`255.255.255.252`
	MLS-GIZA Switch	`Gi1/0/1`	`192.168.102.2`	`255.255.255.252`
	MLS-GIZA VLAN 10 (Mgmt)	`SVI VLAN 10`	`192.168.10.1`	`255.255.255.0`
	PC1	`NIC`	`192.168.10.10`	`255.255.255.0`
	MLS-GIZA VLAN 20 (Users)	`SVI VLAN 20`	`192.168.20.1`	`255.255.255.0`

	PC2	NIC	192.168.20.10	255.255.255.0

4. VLAN Design

Each site runs two VLANs, providing layer-2 segmentation between management infrastructure and end-user workstations. Inter-VLAN routing is performed by the **Multilayer Switch (Layer-3 mode)** using SVIs.

Site	VLAN ID	VLAN Name	Network	Gateway	PCs
Cairo HQ	10	Management	192.168.30.0/24	192.168.30.1	PC3
	20	Users	192.168.40.0/24	192.168.40.1	PC4
Alexandria	10	Management	192.168.50.0/24	192.168.50.1	PC5
	20	Users	192.168.60.0/24	192.168.60.1	PC6
Giza	10	Management	192.168.10.0/24	192.168.10.1	PC1
	20	Users	192.168.20.0/24	192.168.20.1	PC2

5. Routing Strategy

Static Routing with Floating Backup Routes was chosen for this design. Each router has two sets of routes per destination: a **primary route** (default metric) and a **backup/floating route** with **Administrative Distance = 10**, which is installed only when the primary route is removed (simulating link failure).

> Why Static? Simple, predictable, zero overhead – ideal for a fixed 3-site topology. Backup routes with higher AD provide au

5.1 HQ-CAIRO Router Routes

```
ip route 192.168.10.0 255.255.255.0 10.0.0.10         ! Giza VLAN10 via primary
ip route 192.168.20.0 255.255.255.0 10.0.0.10         ! Giza VLAN20 via primary
ip route 192.168.10.0 255.255.255.0 10.0.0.14 10      ! Giza VLAN10 via backup (AD=10)
ip route 192.168.20.0 255.255.255.0 10.0.0.14 10      ! Giza VLAN20 via backup (AD=10)
ip route 192.168.50.0 255.255.255.0 10.0.0.2          ! Alex VLAN10 via primary
ip route 192.168.60.0 255.255.255.0 10.0.0.2          ! Alex VLAN20 via primary
ip route 192.168.50.0 255.255.255.0 10.0.0.6  10      ! Alex VLAN10 via backup (AD=10)
ip route 192.168.60.0 255.255.255.0 10.0.0.6  10      ! Alex VLAN20 via backup (AD=10)
ip route 192.168.30.0 255.255.255.0 192.168.100.2     ! Local VLAN10 via MLS
ip route 192.168.40.0 255.255.255.0 192.168.100.2     ! Local VLAN20 via MLS
```

5.2 GIZA Router Routes

```
ip route 192.168.30.0 255.255.255.0 10.0.0.9          ! Cairo VLAN10 primary
ip route 192.168.40.0 255.255.255.0 10.0.0.9          ! Cairo VLAN20 primary
ip route 192.168.50.0 255.255.255.0 10.0.0.9          ! Alex VLAN10 via HQ primary
ip route 192.168.60.0 255.255.255.0 10.0.0.9          ! Alex VLAN20 via HQ primary
ip route 192.168.30.0 255.255.255.0 10.0.0.13 10      ! Cairo VLAN10 backup
ip route 192.168.40.0 255.255.255.0 10.0.0.13 10      ! Cairo VLAN20 backup
ip route 192.168.50.0 255.255.255.0 10.0.0.13 10      ! Alex via HQ backup
ip route 192.168.60.0 255.255.255.0 10.0.0.13 10      ! Alex via HQ backup
ip route 192.168.10.0 255.255.255.0 192.168.102.2     ! Local VLAN10 via MLS
```

```
ip route 192.168.20.0 255.255.255.0 192.168.102.2     ! Local VLAN20 via MLS
```

5.3 ALEX Router Routes

```
ip route 192.168.10.0 255.255.255.0 10.0.0.1          ! Giza VLAN10 via HQ primary
ip route 192.168.20.0 255.255.255.0 10.0.0.1          ! Giza VLAN20 via HQ primary
ip route 192.168.30.0 255.255.255.0 10.0.0.1          ! Cairo VLAN10 primary
ip route 192.168.40.0 255.255.255.0 10.0.0.1          ! Cairo VLAN20 primary
ip route 192.168.10.0 255.255.255.0 10.0.0.5  10      ! backup AD=10
ip route 192.168.20.0 255.255.255.0 10.0.0.5  10      ! backup AD=10
ip route 192.168.30.0 255.255.255.0 10.0.0.5  10      ! backup AD=10
ip route 192.168.40.0 255.255.255.0 10.0.0.5  10      ! backup AD=10
ip route 192.168.50.0 255.255.255.0 192.168.101.2     ! Local VLAN10 via MLS
ip route 192.168.60.0 255.255.255.0 192.168.101.2     ! Local VLAN20 via MLS
```

6. Python Automation Script

The automation script (**network_automation.py**) reads the current link status from **network_status.json**, evaluates each branch, and automatically activates backup routing when the primary link is detected as DOWN.

6.1 Logic Flow

Step	Action	Output
1	Load network_status.json	Dict with primary/backup link states per site
2	Loop through each site	Site name + link status logged
3	Check primary_link == 'UP'	Route = PRIMARY
4	If primary DOWN → check backup_link	Activate backup, push ip route commands
5	Both DOWN	Route = DOWN, log alert
6	Write final_routes summary	All sites → PRIMARY / BACKUP / DOWN

6.2 Full Script

```
import json
from datetime import datetime
import time

def write_log(message):
    print(message)
    with open("automation_log.txt", "a") as log_file:
        log_file.write(f"[{datetime.now()}] {message}\n")

with open("network_status.json", "r") as file:
    network = json.load(file)

write_log("\n===== NETWORK STATUS =====")
final_routes = {}

for site, data in network.items():
    write_log(f"\nSite : {site}")
    primary = data["primary_link"]
    backup  = data["backup_link"]
    write_log(f"Primary Link Status : {primary}")

    if primary == "UP":
        route = "PRIMARY"
        write_log("Current Route : PRIMARY")
    else:
        if backup == "UP":
            write_log("Backup Link Found")
            write_log("Activating Backup Route...")
            time.sleep(1)
            write_log("Applying Configuration Changes...")
```

```
            write_log("ip route 192.168.50.0 255.255.255.0 10.0.0.5")
            write_log("ip route 192.168.60.0 255.255.255.0 10.0.0.5")
            write_log("Configuration Applied Successfully")
            route = "BACKUP"
            write_log("Current Route : BACKUP")
        else:
            route = "DOWN"
            write_log("No Available Links")

    final_routes[site] = route
    write_log("----------------------------------------")

write_log("\n===== FINAL SUMMARY =====")
for site, route in final_routes.items():
    write_log(f"{site} --> {route}")
```

6.3 JSON Config File (network_status.json)

```
{
    "HQ": {
        "primary_link": "DOWN",
        "backup_link":  "UP"
    },
    "Alexandria": {
        "primary_link": "UP",
        "backup_link":  "UP"
    },
    "Giza": {
        "primary_link": "DOWN",
        "backup_link":  "UP"
    }
}
```

7. Automation Log – Sample Output

The following output was captured from a real execution of the script with HQ primary DOWN (→ BACKUP activated) and Alexandria primary UP.

```
[2026-06-24 23:29:23]
===== NETWORK STATUS =====

Site : HQ
Primary Link Status : DOWN
Backup Link Found
Activating Backup Route...
Applying Configuration Changes...
ip route 192.168.50.0 255.255.255.0 10.0.0.5
ip route 192.168.60.0 255.255.255.0 10.0.0.5
Configuration Applied Successfully
Current Route : BACKUP
----------------------------------------

Site : Alexandria
Primary Link Status : UP
Current Route : PRIMARY
----------------------------------------

Site : Giza
Primary Link Status : UP
Current Route : PRIMARY
----------------------------------------

===== FINAL SUMMARY =====
HQ          --> BACKUP
Alexandria  --> PRIMARY
Giza        --> PRIMARY
```

7.1 Scenario Summary Table

Run	HQ Primary	HQ Route	Alex Primary	Alex Route	Giza Primary	Giza Route
Run 1	UP	PRIMARY	DOWN	BACKUP	UP	PRIMARY
Run 2	DOWN	BACKUP	UP	PRIMARY	UP	PRIMARY
Run 3	DOWN	BACKUP	UP	PRIMARY	UP	PRIMARY
Run 4	DOWN	BACKUP	UP	PRIMARY	DOWN	BACKUP

8. Connectivity Proof

The screenshot below shows a successful cross-branch ping from **PC2 (Giza, 192.168.20.10)** to **PC5 (Alexandria, 192.168.50.10)**, confirming that inter-site routing works correctly across the WAN serial links.

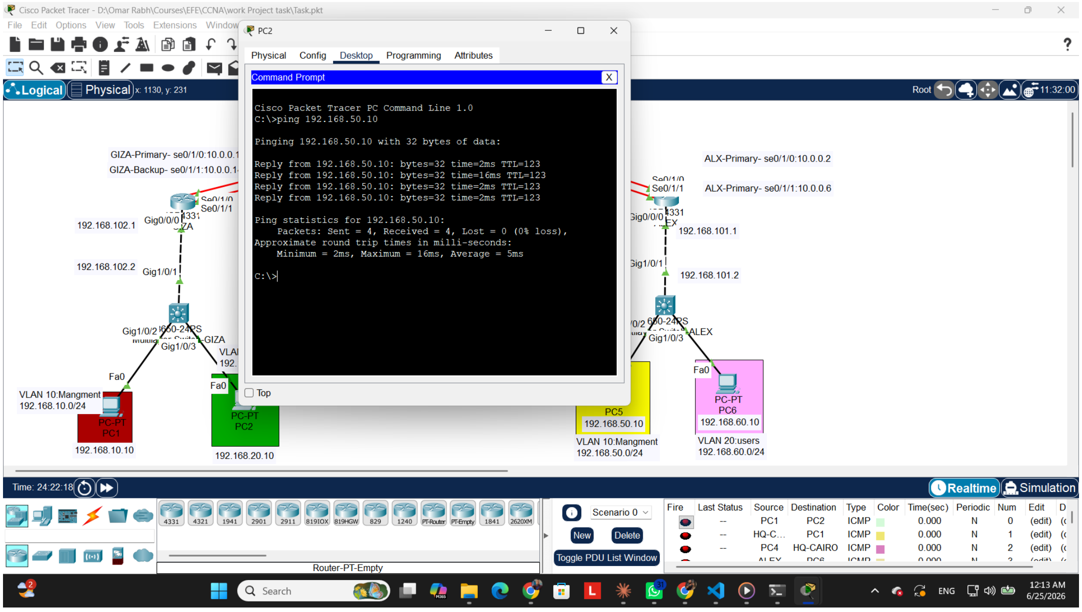

Figure 2 – PC2 (Giza) pinging PC5 (Alexandria) : 4/4 packets received, 0% loss

Ping Result: Sent=4 | Received=4 | Lost=0 (0%) | Min=2ms Avg=5ms Max=16ms ✓

9. Security Considerations

Security Feature	Implementation Detail
SSH v2	All device management uses SSH version 2; Telnet disabled on all VTY lines.
Enable Secret	MD5-hashed enable secret (not plain-text) on all routers and switches.
VLAN Isolation	VLAN 10 (Management) is logically separated from VLAN 20 (Users); inter-VLAN traffic is policy-controlled.
No IP Lookup	'no ip domain-lookup' prevents accidental DNS resolution delays on misconfigured commands.
Banner MOTD	All devices display a legal warning banner on every login.
Backup AD=10	Floating backup routes use AD=10 to ensure they only activate when primary is explicitly removed.
JSON Secrets	In production, credentials and config should reside in a secrets vault, not a plain JSON file.